

UR100040-LQ(EU)

40kW IP65 Liquid Cooling DC Charging Module



Efficient



Silent



Reliable



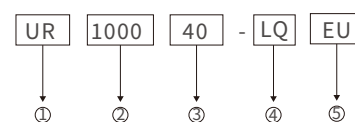
➔ PRODUCT INTRODUCTION

The UR100040-LQ(EU) liquid cooling charging module is designed and manufactured with the world's leading power supply technology and manufacturing technique. The charging module has an ultra-wide voltage range with constant power output. It adopts liquid cooling heat dissipation technology, which greatly improves the heat dissipation efficiency. The liquid cooling charging module has the characteristics of high protection level, high stability, high safety, long service life, high power density, small space occupation, and low operation noise. It has plug-and-pull liquid cooling port and battery back feed protection.

➔ KEY FEATURES

- Ultra-wide output voltage range
- LED panel displaying voltage, current, group, protocol, address, manual/automatic mode and fault information
- Supporting CAN bus communication and software grouping control
- Coolant and circuit separation design, supporting hot swapping, safe and reliable
- IP65 protection level, suitable for various harsh environments
- Industry leading liquid cooling technology solving noise problem permanently
- Peak efficiency > 97%
- Built-in residual voltage discharge circuit
- Standby power consumption < 10W
- Supporting OTA remote upgrading

➔ MODEL DEFINITION



- ① UU Rectifier
- ② Maximum Output Voltage (1000V)
- ③ Rated Output Current (40A)
- ④ Liquid Cooling Heat Dissipation
- ⑤ Europe and American

TECHNICAL PARAMETERS

DC Output

Output voltage	150VDC ~ 1000VDC
Output current	0.5-133.3A
Voltage stabilized accuracy	$\leq \pm 0.5\%$
Current stabilized accuracy	$\leq \pm 1\%$
Voltage ripple	$\leq \pm 1\%$
Output voltage setting error	$\leq \pm 0.5\%$
Efficiency	Peak efficiency>97%

AC Input

Input voltage	260VAC~530VAC
Max. current	<76A
Input frequency	50Hz/60Hz
Power factor	Rated output load PF ≥ 0.99
THDi	<5%
Standby power	<10W

Operating Environment

Operating temperature	-40°C~75°C,derates from 60°C
Storage temperature	-40°C~85°C
Humidity	$\leq 95\%$ RH, without condensation
Pressure/ Altitude	79kPa~110kPa/2000m

Communication

Communication	CAN
HMI	LED operating display
Alarm status	Indicator lights
Number of working modules	Max. 60

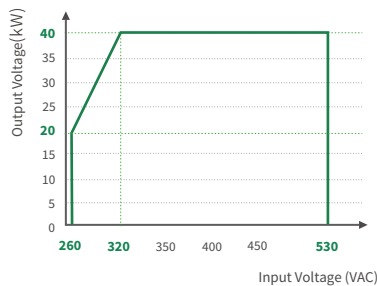
Physical Characteristics

Acoustic noise	≈ 0 dB
Cooling	Liquid
Dimension	300*120*460 (W*H*D/mm)
Weight	<28kg

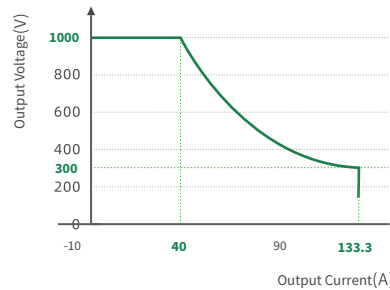
Coolant Parameter

Coolant	6:4 mixture of ethylene glycol and water
Coolant flow rate	6L/min~10L/min
Coolant capacity	1L
Inlet pressure	< 2bar

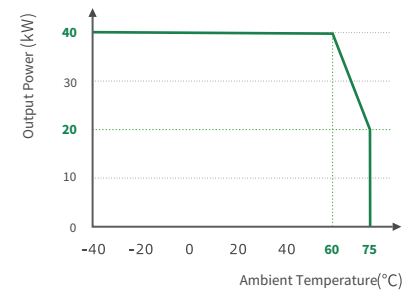
CHARACTERISTIC CURVE



AC Input Characteristic Curve



Output Characteristic Curve ($\leq 60^\circ\text{C}$)



Temperature Characteristic Curve

STRUCTURE SIZE

